Senanayak Sesh Kumar Karri

Research name: K. S. Sesh Kumar

Machine Learning Researcher | Research Fellow, Imperial College London

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Research Summary

Machine-learning researcher at Imperial College London specialising in probabilistic modelling, graph-structured learning, geometry-aware neural networks, and convex and submodular optimisation. PhD in machine learning from INRIA / École Normale Supérieure (SIERRA), followed by research appointments in statistical machine learning and data science. Two papers accepted for poster presentation at NeurIPS 2026, on non-stationary graph Gaussian processes and learning neural-network edge geometry through Banach duality. Research published at NeurIPS, ICML, JMLR, TMLR, and *Machine Learning*. Finance is an application domain for my methodological work: I study learning under non-stationarity, structured high-dimensional estimation, and reinforcement learning for sequential decisions, with applications to financial time series, covariance estimation, and portfolio management.

Accepted Papers at NeurIPS 2026

[1] **K. S. Sesh Kumar** and V. Borovitskiy, **Practical Non-Stationary Graph Gaussian Processes**. *Fortieth Conference on Neural Information Processing Systems (NeurIPS 2026), accepted as a poster*, submission 25512.

[2] **K. S. Sesh Kumar**, **Geometry-Constrained Kolmogorov–Arnold Networks: Learning Edge Geometry via Banach Duality**. *Fortieth Conference on Neural Information Processing Systems (NeurIPS 2026), accepted as a poster*, arXiv:2608.25807.

Academic Positions

Research Fellow (Machine Learning Research) — London, UK
Brevan Howard Centre for Financial Analysis, Imperial College Business School — *Sept 2022 – present*

Research Fellow, Data Science Institute — London, UK
Imperial College London — *Sept 2019 – Aug 2022*

Research Fellow, Statistical Machine Learning Group — London, UK
Imperial College London; Leverhulme Centre for the Future of Intelligence fellowship — *Dec 2017 – Aug 2019*

Postdoctoral Researcher — Klosterneuburg, Austria
Institute of Science and Technology (IST) Austria — *Oct 2016 – Oct 2017*

Education

PhD in Machine Learning, SIERRA, INRIA / École Normale Supérieure — Paris, France
May 2013 – Sept 2016

- Thesis: *On the Links between Probabilistic Graphical Models and Submodular Optimisation*, École Normale Supérieure, 2016.

MSc, Mathematics, Vision, Learning (MVA), ENS Cachan — Cachan, France
Sept 2012 – Apr 2013

B.Tech., Computer Science and Engineering, IIIT Hyderabad — Hyderabad, India
Jul 1999 – Aug 2003

Selected Publications

[1] M. Conserva, M. P. Deisenroth, and **K. S. Sesh Kumar**, **The Graph Cut Kernel for Ranked Data**. *Transactions on Machine Learning Research (TMLR)*, 2022.

[2] C. Buizza, C. Quilodrán Casas, P. Nadler, J. Mack, S. Marrone, Z. Titus, C. Le Cornec, E. Heylen, T. Dur, L. Baca Ruiz, C. Heaney, J. Amador Díaz López, **K. S. Sesh Kumar**, and R. Arcucci, **Data Learning: Integrating Data Assimilation and Machine Learning**. *Journal of Computational Science*, 2021.

[3] R. Moriconi, M. P. Deisenroth, and **K. S. Sesh Kumar**, **High-Dimensional Bayesian Optimization with Manifold Gaussian Processes**. *Machine Learning*, 2020.

[4] **K. S. Sesh Kumar**, F. Bach, and T. Pock, **Fast Decomposable Submodular Function Minimization using Constrained Total Variation**. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[5] R. Moriconi, **K. S. Sesh Kumar**, and M. P. Deisenroth, **High-Dimensional Bayesian Optimization with Projections using Quantile Gaussian Processes**. *Optimization Letters*, 2019.

[6] **K. S. Sesh Kumar** and M. P. Deisenroth, **Differentially Private Empirical Risk Minimization with Sparsity-Inducing Norms**. *Privacy Preserving Machine Learning (PPML) Workshop, NeurIPS*, 2018.

[7] **K. S. Sesh Kumar** and F. Bach, **Active-Set Methods for Submodular Minimization Problems**. *Journal of Machine Learning Research (JMLR)*, 2017.

[8] **K. S. Sesh Kumar** and F. Bach, **Maximizing Submodular Functions using Probabilistic Graphical Models**. *Discrete Optimization for Machine Learning (DISCML) Workshop, NIPS*, 2013.

[9] **K. S. Sesh Kumar** and F. Bach, **Convex Relaxations for Learning Bounded Treewidth Decomposable Graphs**. *International Conference on Machine Learning (ICML)*, 2013.

Preprints and Working Papers

[1] F. Allen, M. T. Kacperczyk, and **K. S. Sesh Kumar**, **Adaptive Market Ecology and Conditional Strategy Performance: Evidence from Machine-Learning Market-Timing Strategies**. *SSRN working paper*, 2026.

[2] F. Allen, M. T. Kacperczyk, and **K. S. Sesh Kumar**, **Modern Machine Learning Tools in Finance: A Critical Perspective**. *SSRN working paper*, 2025.

[3] S. Cohen, **K. S. Sesh Kumar**, and M. P. Deisenroth, **Sliced Multi-Marginal Monge Optimal Transport**. *arXiv:2102.07115*, 2021.

[4] **K. S. Sesh Kumar**, A. Barbero, S. Jegelka, S. Sra, and F. Bach, **Convex Optimization for Parallel Energy Minimization**. *arXiv:1503.01563 / HAL 01123492*, 2015.

Awards and Fellowships

- 2019 — Research Fellowship, Data Science Institute, Imperial College London.
- 2018 — Research Fellowship, Leverhulme Centre for the Future of Intelligence.
- 2017 — Data Science Institute research funding, Imperial College London.

Earlier Research and Industry Experience

Research Engineer, WILLOW team, INRIA, Paris (2012–2013); Research Intern, SIERRA team, INRIA, Paris (2012); Research Intern, Computer Vision and Geometry Group, ETH Zurich (2011); Research Engineer, Télécom SudParis, Evry (2010–2011); GPGPU Engineer, AMD Research and Development, Hyderabad (2006–2010); Research Assistant, CVIT, IIIT Hyderabad (2003–2006), with publications at ICDAR, ICVGIP, DAS, and PReMI.